Cape Capital

Cyborgs are amongst us

Moonshot :
Human hacking – Bionic humans

18 NOVEMBER 2025

Cape Capital

The moonshot

Bionic humans are the convergence of biology and technology to augment, restore, or enhance human physical capabilities sometimes beyond natural limitations.

7f Bionic humans

Human hacking - Medical Equipment

Rank 36

Score 52

Growing / Uncertain

10/11/2025

Definition

Bionic organs and implants restore function for patients with organ failure, amputation, or sensory loss; examples include artificial hearts, advanced limbs, and sensory systems.

Vision

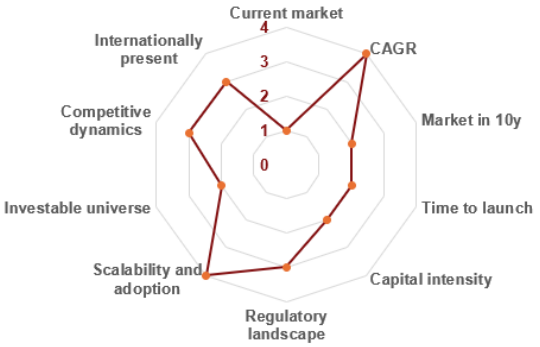
If successful, 10+ million amputees gain AI-powered prosthetics with brain control and sensory feedback by 2035, restoring near-natural function and quality of life; bionic organs extend hope to transplant candidates.

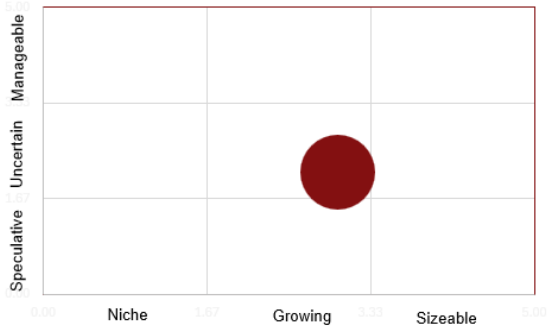
Risk

If progress stalls, amputees remain limited to basic prosthetics with poor function, and artificial organ efforts stall, leaving 500,000+ annual transplant patients without alternatives.

Current market	1	USD 2bn
CAGR	4	31.1%
Market in 10y	**2**	**USD 30bn**
Time to launch	2	Early pilot
Capital intensity	**2**	**High**
Regulatory landscape	3	Moderately supportive
Scalability and adoption	4	Strong adoption drivers
Investable universe	2	
Competitive dynamics	3	
Internationally present	3	

Opportunity	3.00	Growing
Confidence	2.13	Uncertain
Investability	**2.60**	**Medium**





Scoring rationale

We score on clinical evidence for brain-controlled, sensory-feedback prosthetics and artificial organs; durability and biocompatibility; surgical workflow and training; cost and supply readiness; and regulatory and reimbursement milestones. The aim is near-natural performance with safe long-term use.

Convergence (Low-Medium-High)

Next-gen semiconductors (L), Humanoids (L), Physical AI (H), Obesity (GLP1) (M), Applicable longevity / immortality (M), Neurodegenerative diseases (M), Brain-computer interface / Neurotechnology (M), Gene-editing / Synthetic biology (M), New materials (M)

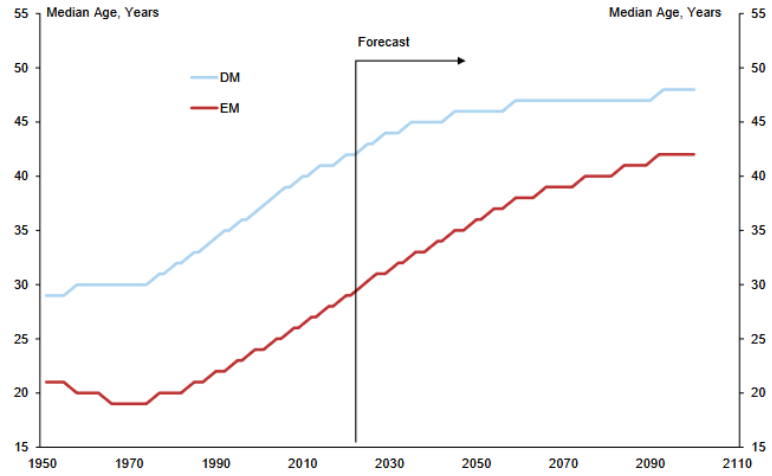
Equities

Cochlear (64.4), DexCom (57.33), Insulet (56.63), Sonova (46.79)

70 is the new 53

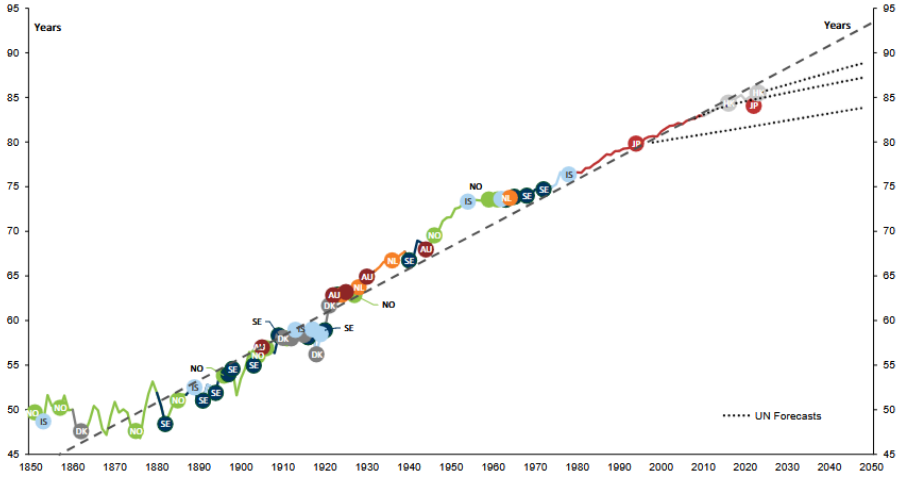
- The global population aged 65+ will reach 1.6 billion by 2050, **driving exponential demand for mobility assistance and rehabilitation technologies**.
- **In 2035, the population aged 65+ will be of 1.16bn** (versus 890m today).

Exhibit 1: In the Past 50 Years, the Median Age Has Risen from 30 to 43 Years in DM Economies and from 19 to 30 in EM Economies
Median Age, Years



Source: United Nations, Goldman Sachs Global Investment Research

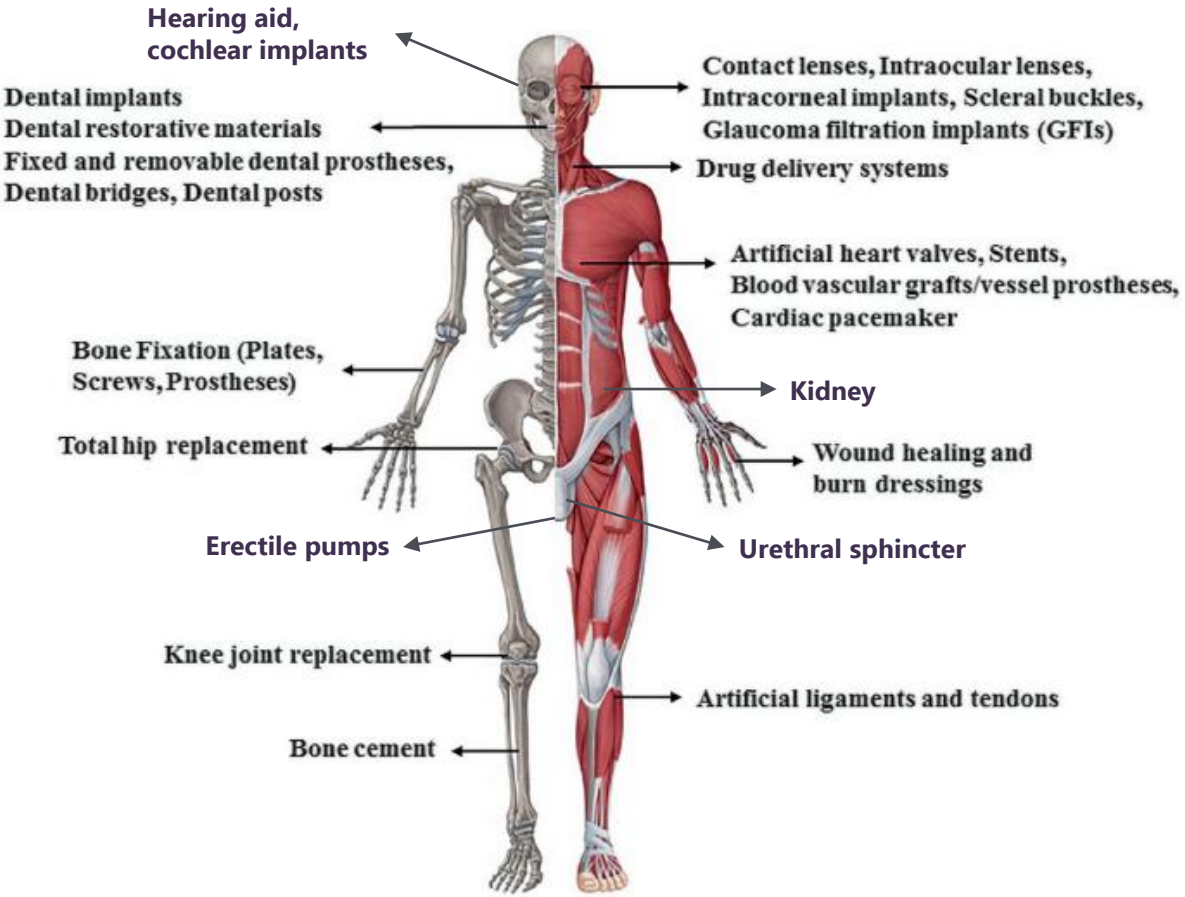
Exhibit 2: Living Longer – Life Expectancy Continues to Chart a Linear Path Higher
Leading country life expectancy (male and female), based on 'period life expectancy' method



By 2033, an est. 50m people globally will use bionic augmentation daily

"We are at the precise moment when the bionic human transitions from science fiction to health insurance—and that regulatory threshold, not technological capability, ***is what unlocks a USD 30 billion market****."*

Larry Jasinski, CEO Lifeward, on Medicare's April 2024 reimbursement decision establishing personal exoskeleton coverage.



The boundary between human and machine is dissolving

In 2025, a patient in Boston controls a prosthetic hand through direct brain-nerve interfaces **while simultaneously feeling texture and temperature through artificial skin**

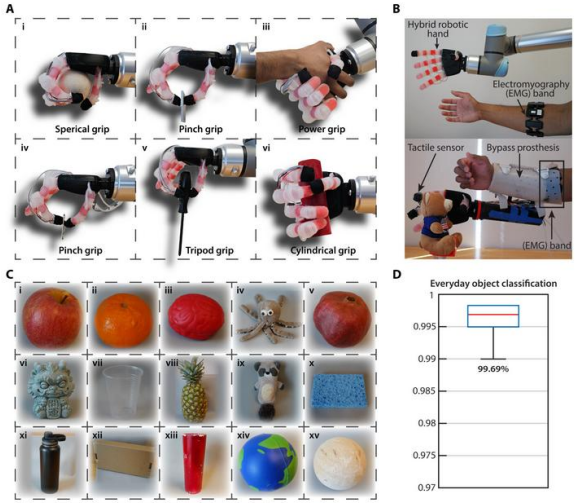
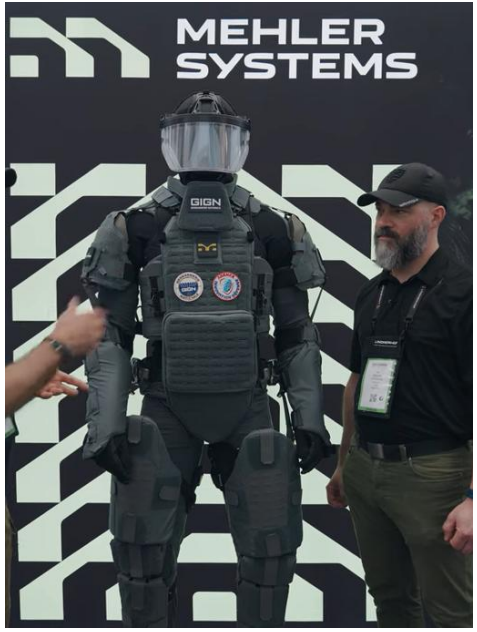


Fig. 6. Hybrid robotic hand grasping and object identification capabilities.
(**A**) Hybrid robotic hand grasping of everyday objects from the SHAP with different grip patterns: (i) spherical grip for a ball, (ii) pinch grip for a heavy 143-g metal plate, (iii) power grip for a handshake with a human hand, (iv) pinch grip for a delicate key, (v) tripod grip for a screwdriver, and (vi) cylindrical grip for a cylinder. (**B**) Hybrid robotic hand controlled by EMG on a bypass prosthesis. (**C**) Fifteen everyday objects that were differentiated using the multilayered tactile sensor of the hybrid robotic hand during grasping. (**D**) The hybrid robotic hand achieved an average classification of 99.69% when grasping and identifying the 15 everyday objects.

Manual work becomes easier, healthier and safer: German Bionic shows how human augmentation is shaping the future of human physical work - Supports up to 38 kg (84 lbs.) per lifting movement

Military grade exoskeleton, providing ballistic protection without additional weight on the operator. The suit alleviates up to 70% of the carry load.



Eyesight, audition

- **Retinal Implants** : The dominant approach places electrodes on or beneath the retina to electrically stimulate surviving neural circuits when photoreceptors have degenerated. Bionic eyes can detect wavelengths invisible to human biology, enabling superhuman vision.
- **Bionic Hearing Technologies** : AI algorithms identify and amplify relevant speech while suppressing background noise—addresses cochlear implant users' primary complaint (difficulty in noisy environments like restaurants)

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INTERVIEW

Un implant rétinien permet à 80 % de patients malvoyants de lire à nouveau

Un essai clinique européen démontre qu'un implant sous-rétinien permet à des patients atteints de dégénérescence maculaire liée à l'âge (DMLA) de retrouver partiellement la vue et de lire à nouveau.

Par Caroline Tourbe

Publié le 20/10/2025 à 14h00

« Aucun implant sous-rétinien n'avait permis à des aveugles de se remettre à lire tout en conservant leur vision périphérique », souligne Serge Picaud, directeur de l'Institut de la vision. © caia image / Alamy Stock Photo / https://www.alamy.com / Alamy Stock Photo

LA SUITE APRÈS CETTE PUBLICITÉ

Artificial Blood – A Critical Frontier in Saving Lives

- The problem : Hemorrhage accounts for ~40% of trauma deaths; **25% of maternal deaths worldwide**.
- The solution : **Must mimics oxygen transport and immune functions of real blood**. Must have extended shelf life, universal compatibility. Must addresses clotting, immune responses, blood pressure.
- Estimated global **artificial blood market valued at USD 11.7 million in 2025**; projected to grow at 20.5% CAGR to USD 62.5 million by 2034. **Broader artificial blood substitutes market estimated at USD 3.7–3.8 billion in 2024**; forecast to USD 8.7 billion by 2035 at 8% CAGR

Japan Launches World's First Clinical Trials for Artificial Blood in 2025

5 months ago 4 min read

Feedback | Share

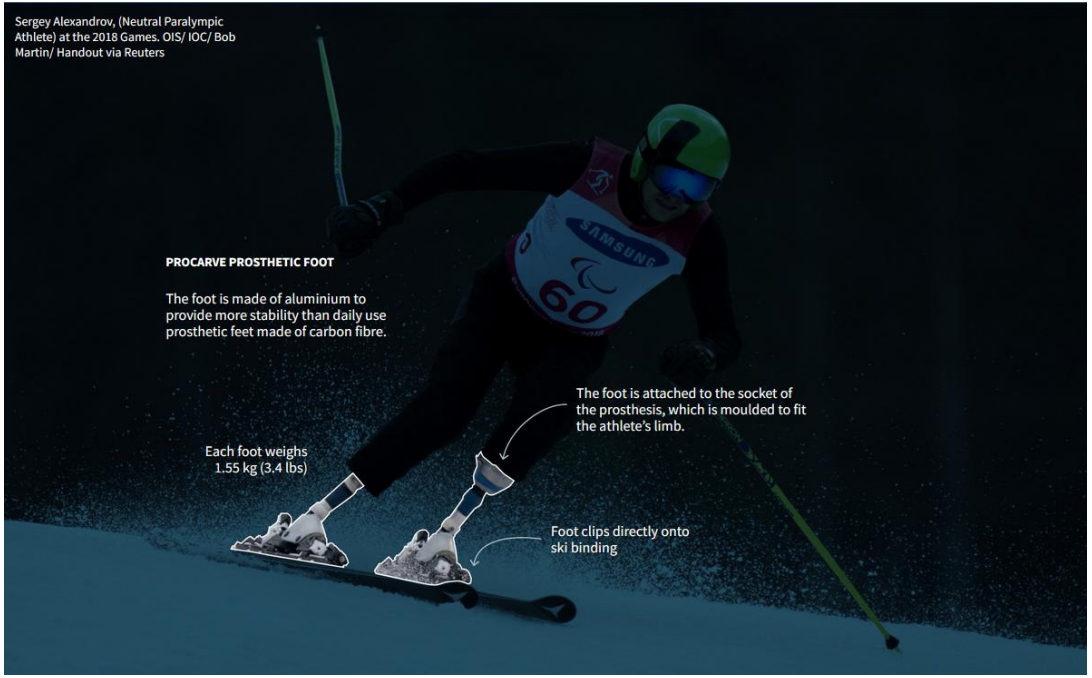
Key Insights

- Japan will begin the world's first clinical trials for artificial blood by March 2025, led by Nara Medical University, marking a potential breakthrough in emergency medicine.
- The artificial blood uses hemoglobin vesicles from expired donations encased in protective shells, eliminating blood type compatibility requirements and offering a two-year shelf life.
- The technology addresses Japan's critical blood shortage crisis driven by an aging population and declining birth rates, with potential global applications.

Japan is poised to make medical history as the first country to begin clinical trials of **artificial blood**, a groundbreaking innovation that could revolutionize emergency medicine and address critical **blood shortage** challenges worldwide. **Nara Medical University** announced that clinical trials involving healthy adults will commence by March 2025, with the ambitious goal of bringing this technology to clinical use by 2030.

Built to win – Para-Olympic games

- The International Paralympic Committee (IPC) makes sure that the equipment used doesn't artificially enhance performance.
- To maintain fair competition, the IPC requires that adaptive equipment is safe, commercially available, and registered for approval well in advance of events.



This article is more than **13 years old**

Oscar Pistorius shouldn't have run in the Olympics – blades are an advantage

Ben Rushgrove

Despite the disappointment of coming sixth in my 100m race, I will never forget the moment the crowd cheered my name

South Africa's Oscar Pistorius starts in the men's 400m semi-final at the London 2012 Olympics. Photograph: Anja Niedringhaus/AP

From medical necessity to elective augmentation

- **Current users are 95% injury/disability-driven**; by 2030, forecasts suggest 25–30% adoption for performance enhancement in sports, military, and industrial contexts.
- **Debate intensifies around "therapeutic vs enhancement" boundaries**, insurance coverage for elective augmentation, and workplace fairness (augmented vs non-augmented workers)

Flex-Run™

Flex-Run with Nike Sole takes distance running to the next level.

Leading-edge bionic devices now exceed biological performance—the Össur Flex-Run **prosthetic generates 30% more energy return than a biological ankle**, enabling Paralympic sprinters to challenge Olympic qualifying times.

ETH Zurich Soft Robotics Lab

***"We're not building better prosthetics—we're growing them**. The difference isn't semantic. It's the difference between manufacturing a machine that breaks and engineering tissue that heals."*

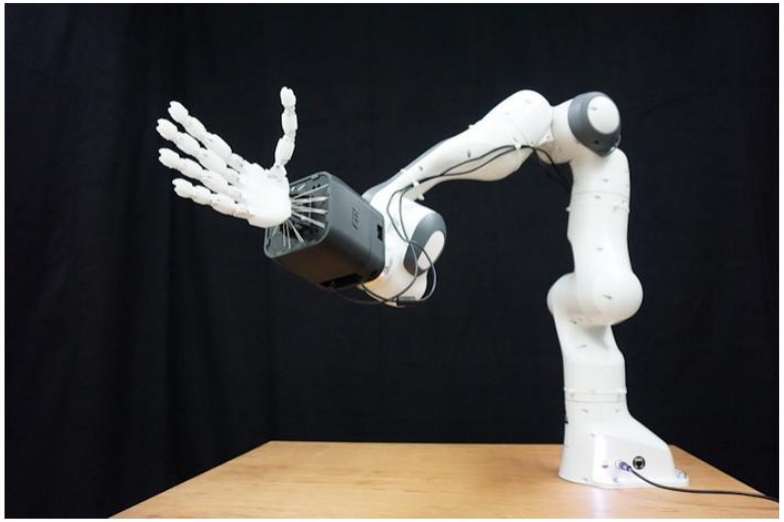
Professor Robert Katzschmann, ETH Zurich, on achieving functional muscle-tendon units in biohybrid actuators

10.09.2025 by Inken De Wit

Read • Share

Muscles are powerful, yet flexible and soft. They also possess a high degree of fine motor control, can grasp even delicate objects, and are capable of self-repair in the case of minor injuries. A key factor in the precise and effective transmission of force is the role of tendons, which connect muscles to bones.

To transfer these properties to robotics, scientists are researching biohybrid solutions. They combine synthetic and biological materials to replicate the structure and function of biological tissue and the movement of living organisms. In the future, biohybrid robots could enhance human-machine interaction in fields such as healthcare, medicine, and assistive robotics. This approach could also advance the development of biohybrid implants for humans.



Robotic arm in the Soft Robotics Lab. (Picture: Soft Robotics Lab/ETH Zurich)

Scorecard web-based (wip)

HUMAN HACKING

Bionic humans

If disruptive bionics land, organ failure becomes reversible; transplant waitlists disappear; critical care becomes manufacturable and distributable. By 2030-35, portable synthetic organs and on-demand blood are clinically validated, ushering in a new paradigm of donorless medicine and dramatically extended healthy lifespan.

45

25

Score (max 65)

31

Convergence score (max 65)

45

20.0% CAGR

USD 30.00bn Est. Future TAM by 2035

Bionic humans

Human hacking

Medical Devices

Score 45

25

Max 65

Convergence 45

31

Max 65

Definition

Disruptive bionics solve fundamental medical bottlenecks by replacing donor-dependent transplantation with manufactured artificial organs. This includes totally artificial hearts, breakthrough synthetic blood, bioengineered lungs, and modular full-organ replacement systems that decouple survival outcomes from donor supply.

Keywords

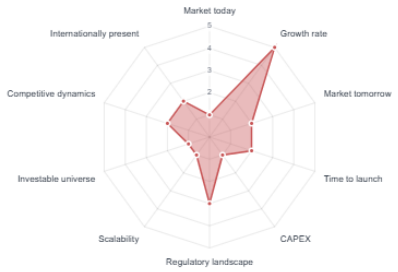
Artificial heart, synthetic blood, bioengineered lungs, total artificial organ, donorless medicine, organ manufacturing, ECMO, perfusion technology, biofabrication, regenerative bionic interface

Risk

Extreme risks: unproven long-term biocompatibility and durability, immune complications, high clinical trial failure rates (Carmat liquidation 2025 is cautionary tale), regulatory complexity, reimbursement resistance, narrow near-term addressable population, possible disruption from xenotransplants or gene therapy. Binary outcomes; winners/losers take all.

Scoring Rationale

Score of 25/60 reflects extreme early-stage disruption with binary outcomes. Carmat collapse is cautionary reminder of capital intensity, regulatory/clinical risk, and market adoption barriers. Upside is massive if disruptive bionics prove durable and safe, but downside includes failure modes. Investable universe is tiny and highly risky. This is pure moonshot thesis requiring 10+ year conviction and tolerance for 80-90% failure rates. Only suited for venture, patient institutional capital, or strategic MedTech partnerships.



Current Market	In 5y Market	Adjusted CAGR	In 10y Market
USD 2.0bn	USD 7.7bn	31.1%	USD 30.0bn

Time to Launch	Capital Intensity	Regulatory	Scalability
Early pilot	Very high capex	Moderate	Difficult to scale

5y Vision

By 2030, first-generation disruptive artificial hearts and synthetic blood approved by FDA/EMA; 500-1,000 patients/year receiving organs; proof-of-concept on durability and safety. Manufacturing begins scaling. Payers reluctantly approve reimbursement after clinical evidence. Large MedTech begins aggressive M&A in space.

10y Vision

By 2035, modular full-body synthetic organ suites reach scale; transplant waitlists essentially eliminated in developed markets. Survival and quality-of-life outcomes exceed biological transplants. Costs drop via industrialization. Ethical paradigm shifts from therapy to augmentation debates.

Convergence

Synthetic biology | advanced materials | AI/ML-driven biofabrication | digital health platforms | gene therapy | immunomodulation

Listed Equity

Abbott Laboratories | Medtronic plc | Terumo Corporation | Abiomed (via J&J) | Getinge AB [all have strategic exposure/partnerships]

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